Evaluation of (probably) obsolete modeling functions

Gabriel Arellano

2026-09-01

Contents

1 Goal

The goal is to make the fgeo2 body of code lighter and easier to maintain by removing obsolete modeling functions. fgeo2 will focus on metrics, descriptors, and other reusable analytical primitives. Statistical modeling requires flexibility and is better handled with maintained, dedicated tools rather than project-specific model-fitting and MCMC infrastructure. This evaluation focuses on:

- primarily, removal of modeling functions;
- minimum edits to the remaining ones; and
- identifying dependents somewhere else, so all code remains working.

2 Candidates for removal

As a starting point, I will use an earlier function-by-function evaluation that was prepared during the transition from the CTFS scripts to the `ctfs` and `fgeo` packages. The table was edited by Sean McMahon and Gabriel Arellano and discussed with Stuart Davies and Mauro Lepore. That table is preserved in `evaluation/old-evaluation-table.xlsx`.

The table below groups the review into three practical categories:

- **Strong:** explicit recommendation to deprecate the complete block, or an explicitly paper-specific implementation marked for deprecation.
- **Weak:** "try to deprecate" or low priority, without a firm recommendation.
- **After checks:** removal appeared appropriate, but the old review suspected that other functions might depend on the code.

The candidates are ordered by module and script so the source is easy to locate. Module `01` means `01_utilities/R/new`; module `05` means `05_demography_growth/R/new`, etc.

Mod.	Script	Function	Old rating	Old evaluation summary
01	utilities-lmerBayes.r	lmerBayes	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	lmerBayes.hyperllike.sigma	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	lmerBayes.hyperllike.mean	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	full.likelihood.lmerBayes	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	llike.fixef.lmer	Strong	Not listed individually; inherits the explicit lmerBayes block recommendation.
01	utilities-lmerBayes.r	llike.model.lmer	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	residual.llike.lmerBayes	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	badSD	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	arrangeParam.llike.2D	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	arrangeParam.Gibbs.2D	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	saveParamFile	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	restoreParamFile	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	resetParam	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	summaryMCMC	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	recalculate.lmerBayesllike	Strong	Explicit recommendation to deprecate the complete lmerBayes block.
01	utilities-lmerBayes.r	covTocorr	Strong	Explicit recommendation to deprecate the complete lmerBayes block.

01	utilities-modelBayes.r	modelBayes	Strong	Explicit recommendation to deprecate the complete modelBayes block.
01	utilities-modelBayes.r	residual.llike.modelBayes	Strong	Explicit recommendation to deprecate the complete modelBayes block.
01	utilities-modelBayes.r	summaryModelMCMC	Strong	Explicit recommendation to deprecate the complete modelBayes block.
01	utilities-statistics.r	regression.Bayes	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	Gibbs.regslope	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	Gibbs.regsigma	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	Gibbs.normalmean	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	Gibbs.normalvar	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	model.xy	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	arrangeParam.llike	After checks	Considered overly specific; check other uses before removal.
01	utilities-statistics.r	arrangeParam.Gibbs	After checks	Considered overly specific; check other uses before removal.
01	utilities-statistics.r	llike.GaussModel	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	llike.GaussModelSD	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	BadParam	After checks	Considered overly specific; check other uses before removal.
01	utilities-statistics.r	graph.modeldiag	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	metrop1step	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	metrop1step.discrete	After checks	The old review suspected other functions might depend on it.
01	utilities-statistics.r	testmcmcfunc	After checks	The old review suspected other functions might depend on it.

05	demogchange-abund.fit.CTFS.r	model.littleR.Gibbs	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	full.abundmodel.llike	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	prob.N1	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	spmean.mort.abundGibbs	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	hyper.abundGibbs	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	hyper.mortGibbs	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	bad.asympower.param	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	bad.asymexp.param	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	fitSeveralAbundModel	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	graph.abundmodel	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-abund.fit.CTFS.r	find.xaxis.hist	Weak	Low-priority exploratory graphic.
05	demogchange-abund.fit.CTFS.r	abundmodel.fit	Strong	Paper-specific model; explicitly recommended for deprecation.
05	demogchange-demogChange.r	lmerMortLinear	Weak	Low-priority model function; no firm deprecation decision recorded.
05	demogchange-demogChange.r	lmerMortFixedTime	Weak	Low-priority model function; no firm deprecation decision recorded.
05	growth-growthfit.bin.r	extract.growthdata	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	run.growthbin.manyspp	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	run.growthfit.bin	Weak	The old recommendation was to try to deprecate the growth-bin block.

05	growth-growthfit.bin.r	growth.flexbin	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	llike.linearbin.optim	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	defineBinBreaks	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	defineSDpar	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	enoughSamplePerBin	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	wideEnoughBins	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	bad.binparam	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	bad.binsdpar	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	calculateBinModel.BIC	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	calculateBinModel.DIC	Weak	Not listed individually; inherits the growth-bin block recommendation.
05	growth-growthfit.bin.r	calculateBinModel.AIC	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	calculateBinModel.bestpred	Weak	The old recommendation was to try to deprecate the growth-bin block.
05	growth-growthfit.bin.r	assembleBinOutput	Weak	Try to deprecate; reconsider binning separately in a more general context.
05	growth-growthfit.graph.r	graph.growthmodel.spp	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	graph.growthmodel	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	overlay.growthbinmodel	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	compare.growthbinmodel	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	graph.outliers.spp	Weak	Low-priority graphic associated with the growth-bin models.

05	growth-growthfit.graph.r	graph.outliers	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	binGraphSampleSpecies	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	binGraphManySpecies.Panel	Weak	Low-priority graphic associated with the growth-bin models.
05	growth-growthfit.graph.r	binGraphManySpecies	Weak	Low-priority graphic associated with the growth-bin models.

Some formal tests indicate that these functions, if conserved, would require maintenance and debugging. In other words: conservation will come with overhead costs.

- The CTFS development notes record that `modelBayes()` still entered debugging and failed after an attempted maintenance fix.
- `growth.flexbin()`, `run.growthfit.bin()`, and `run.growthbin.manyspp()` failed in the growth tutorial. Their attempted calls remain commented out in test-growth_vs_dbh.R.
- The `lmerBayes()` regression test was disabled because its required data were no longer available. Even when enabled, it checked only the names of the returned components, not the fitted result; see test-replace_subset.R.
- The development notes report that one test of `model.littleR.Gibbs()` and `fitSeveralAbundModel()` took approximately 20 minutes. No active automated test for that model remains in the CTFS test suite.

3 Due diligence for dependents and dependencies

This due diligence has two main goals:

1. identify any function outside the candidate set that would break if a candidate were removed; and
2. identify non-candidate functions that may no longer be necessary if all candidates are removed.

This is a static analysis of calls among top-level fgeo2 functions. It does not detect calls constructed with mechanisms such as `get()`, `do.call()`, or `match.fun()`, or uses from tutorials, tests, and external user code. Any apparent absence of dependents therefore remains a prompt for final source and test checks, not automatic authorization to remove code.

3.1 Pre-work snapshot

The dependency structure below uses every R script in the `R/new` directory of all seven fgeo2 modules. The exact pre-work snapshot is commit `040b73e`. Git reads the files directly from that commit, so later changes in the working tree are not mixed into the evaluation.

```
path <- normalizePath(system2("git", c("rev-parse", "--show-toplevel"),
                              stdout = TRUE)[1])

library(igraph, quietly = TRUE)
```

```
Warning: package 'igraph' was built under R version 4.4.3
```

```
Attaching package: 'igraph'


The following objects are masked from 'package:stats':

    decompose, spectrum


The following object is masked from 'package:base':

    union
```

```r
source(file.path(path, "evaluation/0-code-to-source.r"))

pre_work_commit <- "040b73e0cd9a0d55d64c1639b22159d5c6cbd20e"

modules <- c(
  "01_utilities", "02_biomass", "03_spatial",
  "04_abundance_diversity", "05_demography_growth",
  "06_topography_habitat", "07_mapping_plotting"
)

all_files <- system2(
  "git",
  c("-C", shQuote(path), "ls-tree", "-r", "--name-only", pre_work_commit),
  stdout = TRUE
)

module_pattern <- paste(modules, collapse = "|")
scripts <- all_files[
  grepl(paste0("^(", module_pattern, ")/R/new/.*\\.[Rr]$"), all_files)
]

code_env <- new.env(parent = .GlobalEnv)

for(file in scripts)
{
  code <- system2(
    "git",
    c("-C", shQuote(path), "show", paste0(pre_work_commit, ":", file)),
    stdout = TRUE
  )
```

```
  eval(parse(text = paste(code, collapse = "\n")), envir = code_env)
}

objects <- mget(
  ls(code_env, all.names = TRUE),
  envir = code_env,
  inherits = FALSE
)

all_functions <- names(objects)[vapply(objects, is.function, logical(1))]

edges <- get_all_dependencies(envir = code_env)
edges <- edges[, c("from_function", "to_function"), drop = FALSE]
colnames(edges) <- c("from", "to")

candidates <- intersect(candidate_functions, all_functions)
missing_candidates <- setdiff(candidate_functions, all_functions)

if(length(missing_candidates))
  stop("Candidate functions not found: ", paste(missing_candidates, collapse = ", "))
```

3.2 Dependents that could break

For each candidate, the following code follows dependency arrows through all levels and retains only dependents outside the candidate set. These are the functions that would require attention before removal.

```
dependent_rows <- list()

for(candidate in candidates)
{
  direct_dependents <- unique(edges$to[edges$from == candidate])
  direct_dependents <- setdiff(direct_dependents, candidates)

  all_dependents <- downstream_dependents(candidate, edges)
  all_dependents <- setdiff(all_dependents, candidates)

  if(length(all_dependents) == 0)
    next

  dependent_rows[[length(dependent_rows) + 1L]] <- data.frame(
```

```
    Candidate = candidate,
    `Direct dependents outside candidates` = paste(direct_dependents, collapse = ", "),
    `Dependents outside candidates at any level` = paste(all_dependents, collapse = ", "),
    check.names = FALSE
  )
}

if(length(dependent_rows) == 0)
{
  cat("**Result:** No function outside the candidate set depends on a candidate in the pre-wo
} else {
  dependent_table <- do.call(rbind, dependent_rows)
  dependent_table_latex <- knitr::kable(
    dependent_table,
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE,
    row.names = FALSE
  )

  dependent_table_latex <- sub(
    "\\begin{longtable}{lll}",
    "\\begin{longtable}{@{}L{4cm}L{9cm}L{9cm}@{}}",
    dependent_table_latex,
    fixed = TRUE
  )

  cat(
    "\\begin{landscape}\n",
    "\\footnotesize\n",
    "\\setlength{\\tabcolsep}{4pt}\n",
    dependent_table_latex,
    "\n\\end{landscape}\n",
    sep = ""
  )
}
```

Result: No function outside the candidate set depends on a candidate in the pre-work snapshot.

3.3 Dependencies that may become unnecessary

This check follows dependencies upstream through all levels. A non-candidate dependency is marked as potentially unnecessary when it belongs to the candidate dependency chain and has no downstream use outside that potential removal set.

```
candidate_dependencies <- character()

for(candidate in candidates)
  candidate_dependencies <- c(
    candidate_dependencies,
    upstream_dependencies(candidate, edges)
  )

candidate_dependencies <- unique(candidate_dependencies)
candidate_dependencies <- setdiff(candidate_dependencies, candidates)
potential_removal_set <- c(candidates, candidate_dependencies)

dependency_rows <- list()

for(dependency in candidate_dependencies)
{
  all_users <- downstream_dependents(dependency, edges)
  candidate_users <- intersect(all_users, candidates)
  other_users <- setdiff(all_users, potential_removal_set)

  conclusion <- "Still required elsewhere"
  if(length(other_users) == 0)
    conclusion <- "Potentially unnecessary"

  dependency_rows[[length(dependency_rows) + 1L]] <- data.frame(
    Dependency = dependency,
    `Candidate functions downstream` = paste(candidate_users, collapse = ", "),
    `Other downstream functions` = if(length(other_users)) paste(other_users, collapse = ", '
    Conclusion = conclusion,
    check.names = FALSE
  )
}

if(length(dependency_rows) == 0)
{
  cat("**Result:** No non-candidate fgeo2 dependencies were detected.\n")
} else {
```

```
  dependency_table <- do.call(rbind, dependency_rows)
  dependency_table <- dependency_table[
    order(
      dependency_table$Conclusion != "Potentially unnecessary",
      dependency_table$Dependency
    ),
  ]

  dependency_table_latex <- knitr::kable(
    dependency_table,
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE,
    row.names = FALSE
  )

  dependency_table_latex <- sub(
    "\\begin{longtable}{llll}",
    "\\begin{longtable}{@{}L{3.4cm}L{8.1cm}L{8.1cm}L{2.5cm}@{}}",
    dependency_table_latex,
    fixed = TRUE
  )

  cat(
    "\\begin{landscape}\n",
    "\\footnotesize\n",
    "\\setlength{\\tabcolsep}{4pt}\n",
    dependency_table_latex,
    "\n\\end{landscape}\n",
    sep = ""
  )
}
```

Dependency	Candidate functions downstream	Other downstream functions	Conclusion
addBinParam	run.growthfit.bin, run.growthbin.manyspp	None detected	Potentially unnecessary
AssignDiag	lmerBayes	None detected	Potentially unnecessary
CI	model.xy, summaryMCMC, summaryModelMCMC, growth.flexbin, lmerBayes, modelBayes, run.growthfit.bin, run.growthbin.manyspp	None detected	Potentially unnecessary
dasymexp	abundmodel.fit, hyper.abundGibbs, prob.N1, spmean.mort.abundGibbs, full.abundmodel.llike, graph.abundmodel, model.littleR.Gibbs, fitSeveralAbundModel	None detected	Potentially unnecessary
dasymnorm	abundmodel.fit, hyper.abundGibbs, prob.N1, spmean.mort.abundGibbs, full.abundmodel.llike, graph.abundmodel, model.littleR.Gibbs, fitSeveralAbundModel	None detected	Potentially unnecessary
dasympower	abundmodel.fit, hyper.abundGibbs, prob.N1, spmean.mort.abundGibbs, full.abundmodel.llike, graph.abundmodel, model.littleR.Gibbs, fitSeveralAbundModel	None detected	Potentially unnecessary
dsymexp	abundmodel.fit, hyper.abundGibbs, prob.N1, spmean.mort.abundGibbs, full.abundmodel.llike, graph.abundmodel, model.littleR.Gibbs, fitSeveralAbundModel	None detected	Potentially unnecessary
gsp	graph.abundmodel	None detected	Potentially unnecessary
split.data	lmerBayes, restoreParamFile, summaryMCMC	None detected	Potentially unnecessary
assemble.demography	model.littleR.Gibbs, fitSeveralAbundModel	abund.manycensus	Still required elsewhere
attach_if_needed	graph.abundmodel	CTFSplot	Still required elsewhere
colMedians	model.xy, growth.flexbin, run.growthfit.bin, run.growthbin.manyspp	center.predictors, logistic.ctr	Still required elsewhere
convert.factor	model.littleR.Gibbs, fitSeveralAbundModel	abund.manycensus	Still required elsewhere

drp	full.likelihood.lmerBayes, lmerBayes, modelBayes, residual.llike.lmerBayes, residual.llike.modelBayes, summaryMCMC, llike.fixef.lmer, recalculate.lmerBayesllike	angleBisector, constant.linear, growth, ispt.inside, mortality, recruitment, growth.dbh, growth.eachspp, mortality.dbh, mortality.eachspp, recruitment.eachspp	Still required elsewhere
fill.dimension	model.littleR.Gibbs, fitSeveralAbundModel	abundance, biomass.change, biomass.growth, growth, mortality, pop.change.dbh, recruitment, abund.manycensus, abundance.spp, growth.dbh, growth.eachspp, mortality.dbh, mortality.eachspp, recruitment.eachspp	Still required elsewhere
fromjulian	model.littleR.Gibbs, fitSeveralAbundModel	abund.manycensus	Still required elsewhere
IfElse	defineBinBreaks, lmerBayes, metrop1step, model.xy, growth.flexbin, model.littleR.Gibbs, modelBayes, calculateBinModel.bestpred, fitSeveralAbundModel, graph.growthmodel.spp, run.growthfit.bin, binGraphManySpecies, binGraphSampleSpecies, compare.growthbinmodel, graph.growthmodel, overlay.growthbinmodel, run.growthbin.manyspp, binGraphManySpecies.Panel, graph.outliers.spp, graph.outliers	intersection.of.lines, logistic.ctr, logistic.inter, logistic.multiplicative, logistic.power, logistic.power_simple, logistic.standard, parallel.line, angleBisector, constant.linear, distance.to.side, imageJ.to.lxly, intersection.line.curve, line.intersection.pts, logistic.power.mode, pt.to.segment, fullplot.imageJ, logisticmodel.bin	Still required elsewhere
linear.model	lmerMortLinear, calculateBinModel.bestpred, graph.growthmodel.spp, growth.flexbin, binGraphManySpecies, binGraphSampleSpecies, compare.growthbinmodel, graph.growthmodel, overlay.growthbinmodel, run.growthfit.bin, binGraphManySpecies.Panel, graph.outliers.spp, run.growthbin.manyspp, graph.outliers	constant.linear, linear.mortmodel, logisticmodel.bin	Still required elsewhere
linearmodel.bin	calculateBinModel.bestpred, graph.growthmodel.spp, growth.flexbin, binGraphManySpecies, binGraphSampleSpecies, compare.growthbinmodel, graph.growthmodel, overlay.growthbinmodel, run.growthfit.bin, binGraphManySpecies.Panel, graph.outliers.spp, run.growthbin.manyspp, graph.outliers	logisticmodel.bin	Still required elsewhere

linearmodel.bin.set	calculateBinModel.bestpred, graph.growthmodel.spp, growth.flexbin, binGraphManySpecies, binGraphSampleSpecies, compare.growthbinmodel, graph.growthmodel, overlay.growthbinmodel, run.growthfit.bin, binGraphManySpecies.Panel, graph.outliers.spp, run.growthbin.manyspp, graph.outliers	logisticmodel.bin	Still required elsewhere
pop.change	model.littleR.Gibbs, fitSeveralAbundModel	abund.manycensus	Still required elsewhere
pst	compare.growthbinmodel, fitSeveralAbundModel, run.growthbin.manyspp, binGraphManySpecies, binGraphManySpecies.Panel	abund.manycensus, AGB.dbtable, CTFSplot, fullplot.imageJ, get.filename, make.CredIntervalVect, map2species, maptopo, pdf.allplot, regress.loglog, autoregression, biomass.CTFSdb, complete.plotmap, map, png.allplot	Still required elsewhere
pts.to.interceptslope	calculateBinModel.bestpred, graph.growthmodel.spp, growth.flexbin, binGraphManySpecies, binGraphSampleSpecies, compare.growthbinmodel, graph.growthmodel, overlay.growthbinmodel, run.growthfit.bin, binGraphManySpecies.Panel, graph.outliers.spp, run.growthbin.manyspp, graph.outliers	angleBisector, imageJ.to.lxly, intersection.line.curve, line.intersection.pts, pt.to.segment, fullplot.imageJ, logisticmodel.bin	Still required elsewhere
tojulian	model.littleR.Gibbs, fitSeveralAbundModel	create.fulldate, abund.manycensus	Still required elsewhere

4 Current evaluation and decisions

I have reviewed all functions one by one, and in the context of their corresponding working block.

As far as I understand, all this functionality is better covered by standard applications of WinBugs, STAN, `brms` package in R, and other standard tools for Bayesian / hierarchical modeling.

The modeling task itself requires expertise. Experts do know these tools. Overall, it seems very unlikely that any member of the ForestGEO / ATFS communities use (or should use) the code developed by Condit and stored in CTFS.

The recent ForestGEO workshop in Krabi (Thailand) suggests that these tools are becoming more and more accessible and easy to use, even for the average user. The AI revolution will probably make Bayesian statistics even easier to implement.

Some of Condit's code seems legacy code with the partial goal of reproducibility of results, particularly the code in the "abund.fit" block. Although reproducibility is a valuable goal, there are other ways of ensuring that. It does not seem useful that fgeo2 functions as a repository for legacy code.

All points in the same direction: a massive, direct removal of all the functions mentioned above is justified, and it would not have downstream consequences. All code remains under "old" folders as part of the fgeo2 distribution, though.

5 Package structure after removals

The graph is intentionally deferred until after the removals. Once the work is complete, replace the post-work commit and date below and enable the chunk. The graph will use every top-level function in the seven `R/new` directories, including functions with no detected internal dependency edges.

```r
post_work_commit <- "REPLACE_WITH_POST_WORK_COMMIT"
post_work_date <- as.Date("YYYY-MM-DD")

post_all_files <- system2(
  "git",
  c("-C", shQuote(path), "ls-tree", "-r", "--name-only", post_work_commit),
  stdout = TRUE
)

post_scripts <- post_all_files[
```

```r
  grepl(paste0("^(", module_pattern, ")/R/new/.*\\.[Rr]$"), post_all_files)
]

post_code_env <- new.env(parent = .GlobalEnv)

for(file in post_scripts)
{
  code <- system2(
    "git",
    c("-C", shQuote(path), "show", paste0(post_work_commit, ":", file)),
    stdout = TRUE
  )

  eval(parse(text = paste(code, collapse = "\n")), envir = post_code_env)
}

post_objects <- mget(
  ls(post_code_env, all.names = TRUE),
  envir = post_code_env,
  inherits = FALSE
)

post_functions <- names(post_objects)[vapply(post_objects, is.function, logical(1))]

post_edges <- get_all_dependencies(envir = post_code_env)
post_edges <- post_edges[, c("from_function", "to_function"), drop = FALSE]
colnames(post_edges) <- c("from", "to")

post_vertices <- data.frame(name = post_functions)
g.after <- graph_from_data_frame(
  post_edges,
  directed = TRUE,
  vertices = post_vertices
)

remaining_candidates <- intersect(candidate_functions, post_functions)
post_downstream <- character()

for(candidate in remaining_candidates)
  post_downstream <- c(post_downstream, downstream_dependents(candidate, post_edges))

post_downstream <- unique(post_downstream)
```

```r
post_downstream <- setdiff(post_downstream, remaining_candidates)

V(g.after)$color <- "gray80"
V(g.after)$color[V(g.after)$name %in% post_downstream] <- "yellow"
V(g.after)$color[V(g.after)$name %in% remaining_candidates] <- "orange"

post_work_pdf <- file.path(
  path,
  "evaluation/results",
  paste0(
    "fgeo2-dependencies-", substr(post_work_commit, 1, 7), "-",
    format(post_work_date, "%Y-%m-%d"), ".pdf"
  )
)

dir.create(dirname(post_work_pdf), recursive = TRUE, showWarnings = FALSE)

set.seed(1)

pdf(post_work_pdf, height = 33, width = 33)
par(mar = c(0, 0, 2, 0))

plot(
  g.after,
  layout = layout.fruchterman.reingold,
  vertex.label = V(g.after)$name,
  vertex.size = 6,
  vertex.color = V(g.after)$color,
  vertex.frame.color = "white",
  vertex.label.color = "black",
  vertex.label.cex = 2,
  edge.color = "gray45",
  edge.width = 0.7,
  edge.arrow.size = 0.3
)

legend(
  "topleft",
  legend = c(
    "Candidate remaining",
    "Downstream dependent",
    "Other function"
```

```
  ),
  pt.bg = c("orange", "yellow", "gray80"),
  pch = 21,
  pt.cex = 2,
  bty = "n"
)

dev.off()

post_work_pdf_relative <- file.path("results", basename(post_work_pdf))
knitr::include_graphics(post_work_pdf_relative)
```